Preprocessing for transformerCVN and RegCNN training

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Talk with Alejandro

The main task of the RecoEnergyS is to get all the necessary information from the ART-format root files into a f root tree “WC”. This is run with larsoft on the DUNE GPVMs or on the grid as I showed in the large text block fro Sep 1st.

The make_text_file_to_root_trks_shws.C is what actually creates the pixel maps. It creates the images by rebinnin charge depositions along the time and charge axes according to the specified dimensions. I’m sharing the origi version and one I modified to support the atmnu pixel map dimensions. In theory, if the x_coarse and y_coarse variables match what is chosen for the TimeResolution and WireResolution in TransformerCVNMapper.fcl, the p maps should be the same, but I don’t know if anyone checked this.

We then use root2hdf5 to convert the root files to hdf5.

This is where the preprocessing ends for RegCVN. For TransformerCVN, there are a couple more reformatting s at Alex Shmakov’s request.

First, there is this preprocess2.py script, (or preprocess_atmnu.py) I made. Then, there is Alex’s CreateFullySparseDataset.ipynb. This finally creates a single h5 file than can be used to train the network with th training repository. These last steps only involve reformatting and do not change the pixel maps.

Preprocessing workflow:

RecoEnergyS_module.cc

make_text_file_to_root_trks_shws.C

root2hdf5

preprocess2.py

CreateFullySparseDataset.ipynb



RegCNN: red

TransformerCVN: red+blue

Preprocessing for Atmnu by Alejandro on TAU server

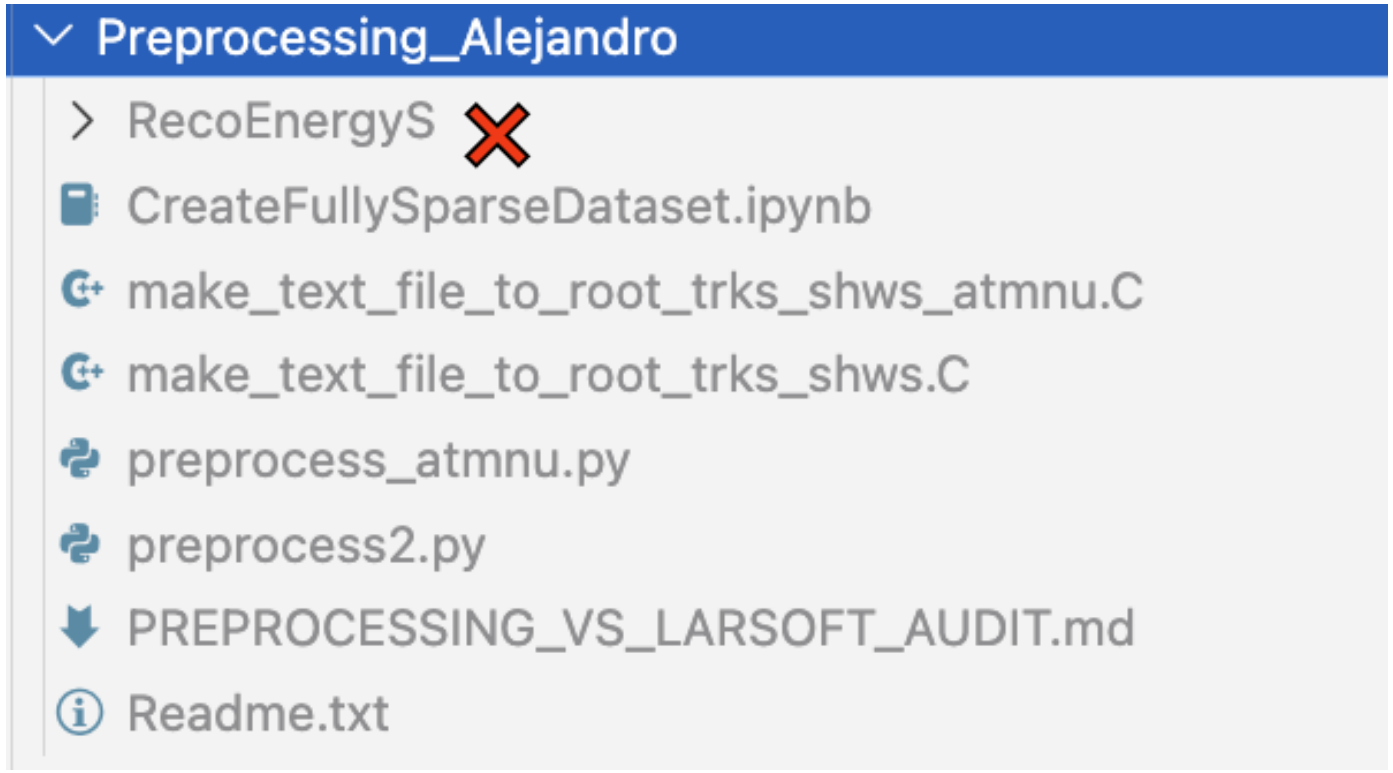
- Dir: /home/ayankelevich/preprocess_dune_atmnu/

```
convert_h5.sh
make_text_file_to_root_trks_shws.C
preprocess_atmnu.py
preprocessed_data_dune_atmnu.test.h5
preprocessed_data_dune_atmnu.train.h5
process_atmnu.sh
```

From recoenergystudies_files
To hdf5_files_for_training

Source codes:

- Dir:/home/binzhang/disk_binzhang/larsoft_test/Preprocessing_Alejandro



- **RecoEnergyS**: /home/binzhang/disk_binzhang/larsoft_test/dune_pixelmap_generator

Conclusions